

REQUIREMENT ANALYSIS DOCUMENT (RAD)

Project Title

Smart Online Shopping System

1. Introduction

The Smart Online Shopping System is an e-commerce application that enables customers to purchase products through the internet. The platform allows users to create accounts, search for products, place orders, make online payments, and track deliveries. Sellers manage product listings and inventory, while administrators supervise the complete system.

The system uses a Relational Database Management System (RDBMS) to store customer, product, payment, and order details securely and efficiently.

2. Problem Statement

Traditional shopping requires customers to visit physical stores, which can be time-consuming and inconvenient. An online shopping platform solves this problem by allowing customers to browse products, compare prices, and purchase items from anywhere at any time.

3. Functional Requirements

Customer Registration

- * New customers can create an account.
- * Existing customers can log in using their email and password.
- * Customers can update their profile information.

Product Browsing

- * View products by category.
- * Search products by name or brand.
- * View product details including price, images, and stock availability.

Shopping Cart

- * Add products to the cart.
- * Update product quantity.

- * Remove unwanted items.
- * View total bill before checkout.

Order Placement

- * Place orders using items in the shopping cart.
- * Receive an Order ID after successful order placement.
- * View order history.

Payment Processing

Supported payment methods include:

- * UPI
- * Credit Card
- * Debit Card
- * Net Banking
- * Cash on Delivery

The system confirms the order only after successful payment (except Cash on Delivery).

Shipment Tracking

Customers can:

- * Track delivery status.
- * View estimated delivery date.
- * Receive notifications when the order is shipped or delivered.

4. Non-Functional Requirements

Performance

- * Fast product search.
- * Quick order processing.
- * Support multiple users simultaneously.

Security

- * User passwords must be encrypted.
- * Secure payment gateway.
- * Role-based login for customers, sellers, and administrators.

Reliability

- * Database backups should be performed regularly.

- * Customer and order data should never be lost.

Availability

- * System should be available 24×7.

5. Actors of the System

Customer

Functions:

- * Register/Login
- * Search products
- * Add to cart
- * Place orders
- * Make payments
- * Track shipments
- * View order history

Seller

Functions:

- * Add products
- * Edit product details
- * Update stock
- * View customer orders
- * Dispatch products

Administrator

Functions:

- * Manage customers
- * Manage sellers
- * Manage products
- * Monitor orders
- * Generate reports
- * Handle complaints

6. Business Rules

- * Every customer must have a unique Customer ID.
- * Every product must have a unique Product ID.
- * Product price cannot be zero or negative.
- * Customers cannot purchase products that are out of stock.
- * Every order receives a unique Order ID.
- * Inventory is automatically reduced after a successful purchase.
- * Each payment is linked to only one order.

7. Reports Required

Product Sales Report

Displays:

- * Product Name
- * Quantity Sold
- * Total Revenue

Customer Order Report

Displays:

- * Customer Name
- * Number of Orders
- * Total Purchase Amount

Inventory Report

Displays:

- * Product Name
- * Available Stock
- * Out-of-Stock Products

Monthly Revenue Report

Displays:

- * Month
- * Total Orders
- * Total Revenue

8. Conclusion

The Smart Online Shopping System provides a secure and efficient platform for online purchasing. It simplifies product management, order processing, payment handling, and shipment tracking while improving the overall shopping experience for customers and helping businesses manage their operations effectively.